

Generative Tectonics: A Dialogue Between Code and Clay

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Abstract

This project, set against the social backdrop of the declining traditional Japanese architectural tile industry, attempts to fuse computational design with traditional ceramic craftsmanship [Martins, 2016]. Using Houdini, we developed a pipeline that procedurally produces complex, non-repeating tile patterns and fabricates them with a custom clay 3D printer. A core innovation is our two-part color system where an AI model generates palettes from user keywords, and a web-based tool applies the Kubelka-Munk model to calculate precise glaze mixtures. This work is a critical inquiry into authorship and ‘making’ in the post-human era, and a deep reflection on the disappearing artisanal knowledge within Japan’s architectural tile industry. The practice is a collaborative negotiation between the artist’s conceptual intention, the machine’s probabilistic logic, the model’s scientific prediction, and the irreducible alchemy of the kiln. This approach reveals how digital technology can serve as a new paradigm for preserving and reactivating tacit knowledge inherent in traditional crafts, contributing to a “Generative Future” where heritage skills are revitalized through technological collaboration.

CCS Concepts

• Applied computing-Arts and humanities-Architecture (buildings);

Keywords

Computational Design, Digital Fabrication, Ceramic 3D Printing, Material Design, Generative AI

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1 Introduction

Recent advances in computational design and digital fabrication have opened new opportunities for rethinking the relationship between code, material, and craft. At the same time, the ceramic tile industry in Japan faces a severe decline, with traditional knowledge and artisanal practices at risk of disappearing [Martins, 2016]. Our research, “Generative Tectonics,” integrates these two trends by uniting computational processes with the contingencies of ceramic materiality, proposing a “Generative Future” where heritage crafts are revitalized through technological collaboration. This work is part of a broader cultural movement in Japan that applies computational methods to ancient traditions, with a prominent precedent being the QUASICRYSTAL project by the traditional Kyoto Nishijin textile company Hosoo, which uses mathematical algorithms to create an aesthetic of “non-periodical repetitions” that could not be achieved by hand.

The central idea of our project—that a set of instructions can become a machine for creating art—finds its intellectual lineage in early conceptual art. Sol LeWitt, a pioneer of this movement, argued that “the idea or concept is the most important aspect of the work”. His Wall Drawing series, executed by assistants according to a predetermined set of instructions, established the precedent for art where the concept, not the artist’s hand, is the source of the work. The project’s use of algorithms and AI is a modern continuation of pioneering generative artists like Harold Cohen, whose AARON software sought to codify the act of drawing [Cohen, 2016]. Similarly, Frieder Nake’s algorithmic art introduced the idea of a “double interpretation” by both human and computer, perfectly describing the negotiation at the heart of our own creative process [Nake, 2018].

2 Methodology and Technical Details

This research demonstrates an integrated workflow that connects procedural modeling, AI-driven glaze design, and clay 3D printing to propose a new methodology for ceramic-based architectural expression. The core concept of “Generative Tectonics” signifies a continuous creative “dialogue” that emerges between the logic of abstract “code” and the physical unpredictability of “clay”. This dialogue explores the “tacit knowledge” of the craftsperson—a knowledge that is often ineffable and embodied, and which a machine-driven process can help to articulate and preserve.

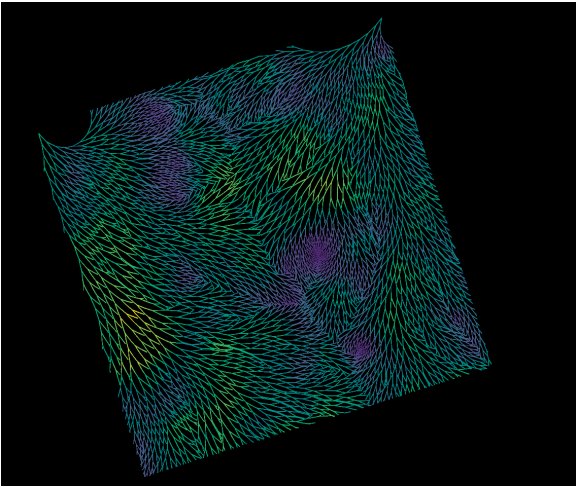


Figure 1: Procedural modeling of ceramic tile patterns in Houdini.

This dialogue is defined as a creative tension between precisely controlled digital processes and the material’s inherent unpredictability. The chemical and physical changes that occur within the kiln are creative forces beyond the artist’s control, symbolizing a natural “serendipity” that complements the logical process of human and machine. This productive “noise” or “error” is central to the work’s philosophy, aligning with works such as *Searching All Sources of White*, which presents “the error as the landscape” [Hui, 2020].

The workflow begins with Houdini’s procedural capabilities to generate complex tectonic patterns that would be difficult to model manually. An algorithm recursively branches surface vectors to create organic patterns across the tile’s surface. The generated geometry is then segmented into printable units and automatically converted into G-code, demonstrating how procedural design can directly inform ceramic production.

For color, a two-stage system was developed to overcome the trial-and-error nature of traditional glaze design. The first stage uses an AI model to generate context-aware color palettes from natural language keywords. The second stage applies the Kubelka-Munk (KM) model, a physics-based method that predicts reflectance from pigment absorption and scattering coefficients, to estimate the mixing ratios needed to reproduce target colors. The generated models are fabricated using a screw-based clay 3D printer and then fired at 1,200°C.

3 Conclusion

This research presents an integrated workflow that connects procedural modeling, AI-driven glaze design, and clay 3D printing to propose a new methodology for ceramic-based architectural expression. By bridging computational control with material serendipity, the system establishes a framework where abstract digital intent is precisely, yet unpredictably, translated into tangible outcomes.

The installation is designed for an immersive experience, inviting the audience to observe the details of the tiles, the gloss of the glaze

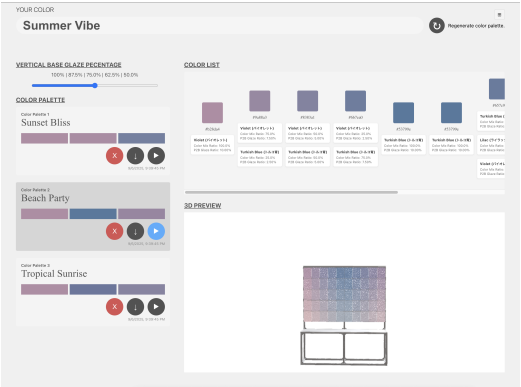


Figure 2: Glaze mixing simulator with AI assistant.



Figure 3: Fabricated clay 3D printed tiles with gradient glaze application.

samples, and the production process. The work’s core philosophical contribution can be tied to projects like *Cerebral Interaction and Painting*, which used a user’s mind to interact with a digital painting to explore the philosophical concept of Qi [Huang and Lioret, 2013].

Ultimately, the questions this project poses are multi-faceted: “What does it mean to be a craftsman when your primary tool is an algorithm?”, “Where does the hand’s knowledge reside in a world of digital fabrication?”, and “How can we re-establish a living archive for skills on the brink of extinction?”. This installation is our answer: a wall that is both a product of code and a tribute to the enduring, beautiful unpredictability of the physical, capturing a moment in a continuous, “generative becoming.”

A key to the project’s future is the establishment of a closed-loop system. We propose a future iteration where real-world data from the kiln—such as precise color discrepancies and physical distortions from firing—is fed back into the AI and Kubelka-Munk models. This would allow the system to learn from the material’s unpredictable nature, making it a truly symbiotic and continuously evolving creative partner.

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